DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 3

Data engineer

Profession 3 of 10 in Development and IT.

What the profession means

A data engineer builds the pipelines that collect, clean, store and serve data so analysts and models can rely on it.

What professionals do every day

Designs data pipelines, writes transformation code, schedules jobs, monitors failures, manages storage and documents the data.

Why businesses hire them, and what problems they solve

Analysis is only as good as its data. Engineers make sure the numbers arrive on time and mean what they say.

- Late or broken reports.
- Duplicate or inconsistent records.
- Pipelines that are impossible to maintain.

Skills you need

- SQL and Python.
- ETL and pipeline scheduling.
- Data modelling and warehousing.
- Monitoring and data quality checks.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
Apache Airflow	Pipeline scheduling	Free (self-hosted)	Managed services vary	airflow.apache.org
dbt	Transformations in SQL (verify plans)	Free core	Paid cloud	www.getdbt.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Load a CSV and clean it with a repeatable script.
Intermediate	Build a scheduled job that loads one table.

Advanced	Add data quality checks and an alert.

Projects by level

- **Beginner:** Sales pipeline for a shop.
- **Intermediate:** Order-to-report pipeline for a firm.
- **Advanced:** Multi-source warehouse for a clinic.

Client-ready portfolio project

A small pipeline: load, clean, check and publish a table, with a README, tests and a failure alert.

Worked example

Worked example. A daily load rejects rows with a negative amount and logs their ids; the report uses only validated rows.

Assignment with full solution

ASSIGNMENT

Data quality check

Business problem: A shop's order file sometimes contains bad amounts.

Steps

1. Write the SQL check that finds missing or non-positive amounts.
2. Run it on the sample data and record any rows found.
3. Add the check to the daily job and send an alert when it finds rows.

Expected output: The check, the result and the alert rule.

SOLUTION (ANSWER KEY)

The sample data contains no missing amounts, so the check returns no rows. A check that returns no rows is still a passing check; the alert triggers only when rows are found.

SQL

```sql
-- data quality check: orders with a missing or non-positive amount
SELECT id, amount FROM orders WHERE amount IS NULL OR amount <= 0;
```

Common mistakes

- Pipelines without tests.
- Hard-coded paths and secrets.
- No alert on failure.
- Overwriting history without a record.

Freelancing opportunities

- Data pipeline builds for small firms.
- Maintenance contracts for data jobs.
- Briefs for 'data pipeline', 'ETL', 'Airflow', 'dbt'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Apache Airflow documentation
- dbt documentation

Part of **Master Freelancing: Data Analytics and Data Science**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.